

Sub-wavelength plasmonic modes in a conductor-gap-dielectric system with a nanoscale gap

Ivan Avrutsky^{1,*}, Richard Soref² and Walter Buchwald²

¹Department of Electrical and Computer Engineering, Wayne State University, Detroit, MI 48202, USA

²Sensors Directorate, Air Force Research Laboratory, AFRL/RHHC, Hanscom Air Force Base, Massachusetts 01731-2909, USA

*ivan.avrutsky@wayne.edu

Abstract: We study guided modes in a conductor-gap-dielectric (CGD) system that includes a low-index dielectric gap layer of deep sub-wavelength thickness sandwiched between a conductor and a high-index dielectric cladding. Analysis of the dispersion equation for CGD modes provides an analytical estimation for the cut-off thickness of the gap layer. This guided mode is unusual because it exists when the gap thickness is less than the cutoff thickness. In the direction normal to the interfaces, the modal electric field is tightly confined within the gap. Sub-wavelength lateral mode confinement is readily provided by a spatial variation of the gap-layer thickness: the modal field localizes at the narrowest gap. Various lateral confinement schemes are proposed and verified by numerical simulations. Possible applications of CGD modes include surface-plasmon nano-lasers (SPASERS) and sensors. If these plasmonic waveguides are scaled for operation at far infrared rather than telecom wavelengths, then the propagation losses are dramatically reduced, thereby enabling the construction of practical chip-scale plasmonic integrated circuits or PLICs.

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1. Introduction

Confining electromagnetic waves to a deep sub-wavelength domain has been a topic of intense research in recent years. Mode confinement below the optical diffraction limit – the subject of this paper – is interesting not only from an academic standpoint, but also motivates the development of novel plasmonic devices substantially more compact than their photonic predecessors. Infrared confinement at the deep sub-wavelength level has value for enabling "point like" light emitters and densely integrated on-chip waveguide systems.

The main goal of this paper is to investigate new plasmonic waveguide modes possessing extreme confinement. The secondary goal is to determine whether practical plasmonic structures can be constructed using the undoped semiconductors Ge and Si as "dielectrics" in

the plasmonic device. In other words, we are focusing upon CMOS-compatible (manufacturable) “group IV plasmonics”. Regarding the plasmonic “conductors” considered in this paper, they could be any materials that exhibit negative dielectric permittivity in the IR frequency range of interest. Thus the conductors can be metals or the various silicides (Pd₂Si, PtSi, NiSi, etc) whose complex permittivities are quite similar to those of metals.

2. Waveguide techniques for strong mode confinement

For dielectric waveguides, a stronger confinement is achieved with a larger index step between the core and cladding [1] – [2]. Optical waveguides of this type are also called photonic wires. The best example is a Si (refractive index $n_{Si} = 3.48$) core waveguide with silica ($n_{SiO_2} = 1.44$) or even vacuum ($n_v = 1.0$) claddings. Direct calculation shows that with 3.48/1.0 index contrast, a slab as thin as $\lambda/20$, where λ is the vacuum wavelength of the guided wave, provides confinement of a transverse electric (TE) mode to a region as narrow as $\lambda/10$ as judged by the full width at half maximum (FWHM) of the mode’s squared electric field distribution. In practice, to reduce possible light scattering at the interfaces, the waveguides are often much thicker, typically around $0.15\lambda - 0.20\lambda$, and the claddings is made of silica (index contrast 3.48/1.44). The width of the distribution of electric field squared in this case is in the range $0.13\lambda - 0.15\lambda$. In a two-dimensional square $0.2\lambda \times 0.2\lambda$ Si core waveguide surrounded by silica cladding a guided mode with electric field mainly parallel to a side of the core cross section is confined to approximately $0.11\lambda \times 0.15\lambda$ at the half-intensity level. Any new waveguide structure claimed to provide stronger light confinement should be able to keep light waves within an even smaller cross section.

Dielectric slot waveguides have been proposed recently to achieve stronger field confinement. Many different structures have been considered addressing the issue of light confinement in two dimensions [3] – [7], but the simplest example of a one-dimensional structure that possesses the required property is a dielectric slab waveguide supporting a transverse magnetic (TM) mode modified by inserting a low-index dielectric layer – the slot – in the middle of the slab. If the slot is narrow enough, the mode profile remains about the same as it was in the slab without the slot, except for the slot area itself. Here, due to the continuity of the electric displacement vector at the boundaries, the normal component of the electric field in the slot is increased by a factor of $(n_w/n_s)^2$, where n_w is the index of the guiding slab, and n_s is the index of the material in the slot. The buildup of the field in a low-index slot layer is due to surface polarization charges that appear at the surface of a material with a large value of dielectric permittivity. The factor $(n_w/n_s)^2$ can be as large as large as 12.1 for $n_w/n_s = 3.48/1$ or 5.8 for $n_w/n_s = 3.48/1.44$. The full width at half maximum for the guided mode of a slot waveguide is as narrow as the slot size, and can easily go down to single nanometers. However, as long as the field distribution is no longer a smooth bell-like curve, the full width at half maximum would be a bad metric because it ignores the fact that there may be just a tiny fraction of total mode power carried within the slot region. A more appropriate parameter is the effective mode size defined as the maximal light intensity divided by the total power carried by the mode.

Since wavelength is the only length-scale parameter in classical electrodynamics, researchers tried to apply radio-frequency solutions to optics. For example, a coaxial cable with a few millimeters cross section perfectly confines an electromagnetic wave with a vacuum wavelength of meters or longer. Applied to optical frequencies, this approach would allow for nanoscale coaxial cables to guide light waves [8] – [9]. In a coaxial cable, the central conductor placed inside of the outer cylindrical conductor needs to be thick enough to remove the singularity associated with cylindrical symmetry. When it comes to optical frequencies, the finite penetration depth into metal sets the limit for the size of the internal conductor. Further, to keep the losses low, the cross section of the dielectric part of such a cable must be much larger than the part of the conductor’s cross section penetrated by the light waves. This eventually defines the cross section of the entire “conductive optical cable”, which is not

much smaller than the cross section of a dielectric waveguide. Because of this, special plasmonic based waveguide modes offer a better approach.

Surface plasmon polaritons and phonon polaritons at the interface between a conductor and a dielectric medium have electromagnetic field decaying exponentially away from the interface and thus provide strong confinement [10] – [12]. A thin film of conductor located between dielectric claddings supports hybrids of the surface modes known as short-range and long-range plasmonic modes [13] – [14]. Plasmonic modes at non-flat interfaces have also been studied including metallic wires [15] – [17], V-grooves [18] and ridges [19]. In general, the EM field “wings” extending into the dielectric are rather long for all these modes, except, perhaps, for the case of the so called resonant surface plasmons [20] – [21].

Extremely strong light confinement is found in a dielectric gap layer located between two conductors [22] – [23]. These modes are commonly referred to as gap plasmon polaritons or gap phonon polaritons. Theoretically, a guided mode in such a structure exists for an infinitely narrow gap layer, while field penetration into the conductors is limited and defined by the conductors’ optical constants. The problem with the gap modes is that stronger confinement is accompanied by large propagation losses. In particular, when gap shrinks to zero, losses become infinitely large.

3. New gap-plasmon modes

Guided modes with very strong light confinement have been recently discovered in a system composed of a high-index dielectric cylinder separated from a metal surface by a narrow low-index dielectric gap [24] – [26]. It has been shown that the modes in such a structure can be tightly confined in the vicinity of the gap, which can be as narrow as few nanometers for light with vacuum wavelength in near infrared. Excellent experimental evidence has been produced by demonstrating lasing at the wavelength of 490 nm in CdS nanowires separated from an Ag surface by a 5-nm-thick MgF_2 dielectric gap layer. These modes have been called Hybrid Plasmon Polaritons (HPP) as their origin was believed to be linked to hybridization (that is, coupling) of surface plasmon polariton (SPP) modes and the modes of a dielectric cylinder [25] – [26] (or of a dielectric waveguide with other, e.g., rectangular cross section [24]). Numerical simulation performed with a commercial mode solver has been reported indicating a tightly confined mode with strong electromagnetic field in the gap and some penetration into the high-index dielectric cylinder. In this paper, we present evidence that the hybrid hypothesis is not accurate.

In this paper, we study guided modes in a low-index gap layer sandwiched between a conductor from one side and a high-index dielectric from the other side – the system herein referred to as conductor-gap-dielectric, CGD. The structure is schematically shown in Fig. 1. We analyze the dispersion equation, find the mode cutoff thickness of the gap layer, provide numerical examples showing the modal index and losses in various structures, and suggest possible approaches for lateral confinement of the CGD modes. To the best of our knowledge, this is the first report on guided modes in a CGD system.

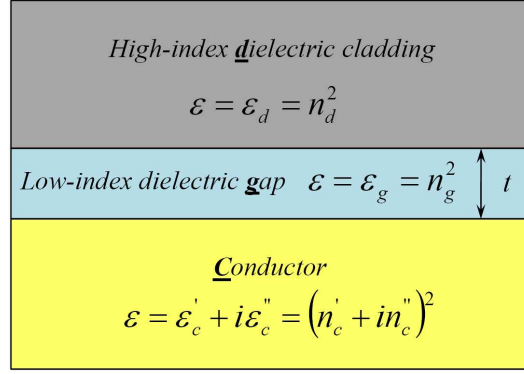


Fig. 1. Schematic view of the conductor-gap-dielectric (CGD) structure comprising a low-index dielectric gap layer between high-index dielectric and conductor claddings.

Careful examination of the numerical data related to hybrid plasmon polaritons [24–26] reveals that the light confinement takes place in the close vicinity of the gap so that it is unlikely that the modal structure of the high-index dielectric above the gap has substantial influence on the gap mode. While the hybridization concept looks reasonable, a few things do not quite match this explanation. First, the structure with a nano-scale gap may support only one mode, while the hybridization in its pure form requires two modes, and then two, not one, hybrid modes would appear as linear combinations of individual modes. Second, it remains mysterious as to why the gap should be in the single nanometers range, very much smaller than the extent of the surface plasmon polariton and the guided mode of the dielectric cylinder.

A more careful examination of the modes in a low-index gap between a high-index dielectric and a conductor shows that these modes are an intrinsic property of the gap structure – and they are the very subject of our study. The guided mode formed by the gap structure is not really sensitive to the exact geometry of the high-index part of the system far from the gap. In fact, the high-index material can be extended infinitely far from the gap and thus supporting no modes of its own, however, the gap mode is still there. A uniform gap size does not give lateral confinement but a spatially varying gap does. As the thickness of the low-index gap layer increases, it effectively reduces the modal index of the plasmonic mode in that thicker gap which in turn provides confinement in the lateral direction. This is the only reason why the cylindrical geometry is helpful. It does not have to be a cylinder, though. It could be, for example, a parabolic tip approaching the metal surface, or a structure of some other shape.

This does not diminish in any sense the significance of the theoretical predictions published in [24] – [25] and experimental demonstration in [26]. We would like to emphasize, however, that referring to the hybridization and coupling of a surface plasmon polariton to a mode of a cylinder is misleading when describing the modes with strongest confinement. The gap structure would work equally well for a greatly diverse set of geometries as long as it has the right gap size and appropriate lateral gap size variation to produce lateral confinement. We consider it important to provide this clarification because it eliminates the requirement of forming an actual high-index dielectric waveguide. This in turn opens more opportunities for experimental studies of the gap modes.

4. Theory of guided modes in a conductor-gap-dielectric system

The basic structure under analysis is a layer of low-index dielectric gap with refractive index $n_g = \sqrt{\epsilon_g}$ and thickness t sandwiched between a semi-infinite conductor with dielectric permittivity $\epsilon_c = \epsilon'_c + i\epsilon''_c$ and semi-infinite dielectric cladding with refractive index

$n_d = \sqrt{\varepsilon_d} > n_g$, as shown in Fig. 1. In the absence of the gap, $t = 0$, the structure is reduced to an interface between a dielectric and conductor, which is known to support a surface plasmon polariton (or surface phonon polariton, depending on the physical origin of negative real part of ε_c) with modal index given by

$$n_{SPP} = \sqrt{\frac{\varepsilon_d \varepsilon_c}{\varepsilon_d + \varepsilon_c}}. \quad (1)$$

For TM polarized waves, the confined mode exists when $\varepsilon_d + \varepsilon_c' < 0$, and, in general, the SPP modal index will be higher than the refractive index of the dielectric cladding $Re(n_{SPP}) > n_d$. Within the approximation of a “good conductor” ($\varepsilon_c' < 0$, $\varepsilon_c'' \ll |\varepsilon_c'|$, and $\varepsilon_d \ll |\varepsilon_c'|$), the modal index of SPP is then found via a Taylor expansion of (1). The difference $\Delta n_{SPP} = Re(n_{SPP}) - n_d$ is estimated to be

$$\Delta n_{SPP} = Re(n_{SPP}) - n_d \approx \frac{n_d^3}{2|\varepsilon_c'|}. \quad (2)$$

The index difference Δn_{SPP} is approximately proportional to the third power of the index of the dielectric cladding, and thus can reach relatively high values for high-index dielectrics. Propagation losses of SPPs $\alpha_{SPP} = (4\pi/\lambda)Im(n_{SPP})$, scale with the cladding index in a similar way:

$$\alpha_{SPP} \approx \frac{2\pi}{\lambda} \frac{n_d^3 \varepsilon_c''}{\varepsilon_c'^2}. \quad (3)$$

Equation (3) also shows that, in terms of the $1/e$ intensity propagation length of SPPs, the figure-of-merit for good conductors is $\varepsilon_c'^2 / \varepsilon_c''$, which helps to select suitable materials.

The field penetration depth is found using (2). Taking the field strength at the interface as a reference, in the direction normal to the interface the field in dielectric cladding is reduced by a factor of e at the distance

$$l_d = \frac{\lambda}{2\pi} \frac{1}{Re(\sqrt{n_{SPP}^2 - n_d^2})} \approx \frac{\lambda}{2\pi} \frac{\sqrt{|\varepsilon_c'|}}{n_d^2}. \quad (4)$$

Given the permittivity of conductor, the losses for the surface plasmon polariton are directly proportional to the third power of the dielectric index (3), while penetration depth is inversely proportional to the second power of refractive index of dielectric (4).

The field penetration depth into the conductor cladding in the direction normal to the interface is estimated to be

$$l_c = \frac{\lambda}{2\pi} \frac{1}{Re(\sqrt{n_{SPP}^2 - \varepsilon_c})} \approx \frac{\lambda}{2\pi} \frac{1}{\sqrt{|\varepsilon_c'|}}, \quad (5)$$

and, to a great degree, is independent of the optical properties of the dielectric cladding as long as $\varepsilon_d \ll |\varepsilon_c'|$.

It follows from (2) – (5), that, in the case of high-index materials, the traditional SPP mode found at the interface between a conductor and a dielectric, may have a modal index substantially higher than the refractive index of the dielectric ($Re(n_{SPP}) - n_d \sim n_d^3$,

propagation losses may be high $\alpha_{SPP} \sim n_d^3$, and the plasmon's field will be strongly confined to the surface $l_d \sim 1/n_d^2$ of the conductor.

We now introduce a low-index gap layer of thickness t between the high-index dielectric and conductor claddings to form the conductor-gap-dielectric (CGD) structure of Fig. 1. By representing the fields in the gap structure as combinations of plane waves and by applying appropriate boundary conditions, the dispersion equation for the modes in the gap structure is found to be as follows

$$e^{-2qt} \left(\frac{q}{\epsilon_g} - \frac{p}{\epsilon_d} \right) \left(\frac{q}{\epsilon_g} - \frac{r}{\epsilon_c} \right) = \left(\frac{q}{\epsilon_g} + \frac{p}{\epsilon_d} \right) \left(\frac{q}{\epsilon_g} + \frac{r}{\epsilon_c} \right), \quad (6)$$

where

$$p = \frac{2\pi}{\lambda} \sqrt{n_{CGD}^2 - \epsilon_d}, \quad q = \frac{2\pi}{\lambda} \sqrt{n_{CGD}^2 - \epsilon_g}, \quad r = \frac{2\pi}{\lambda} \sqrt{n_{CGD}^2 - \epsilon_c}. \quad (7)$$

and n_{CGD} is the modal index of the CGD waveguide. With appropriate definition of complex-valued functions (7), and suitable choice of material constants, this dispersion equation is mathematically equivalent to the dispersion equations for TM modes in any three-layer system including the dielectric slab waveguide, gap plasmon polaritons in a thin dielectric gap between metallic claddings, and the short-range and long-range plasmonic modes of a thin metal film. We choose to write it in the particular form of (6) and (7) because it is more suitable for analytical analysis of modes on the CGD structure. In addition, the validity of Eqs. (6) and (7) are confirmed by considering the case of $t = 0$ leading to $n_{CGD}|_{t=0} \equiv n_{SPP} = \sqrt{\epsilon_d \epsilon_c / (\epsilon_d + \epsilon_c)}$ which is in perfect agreement with Eq. (1).

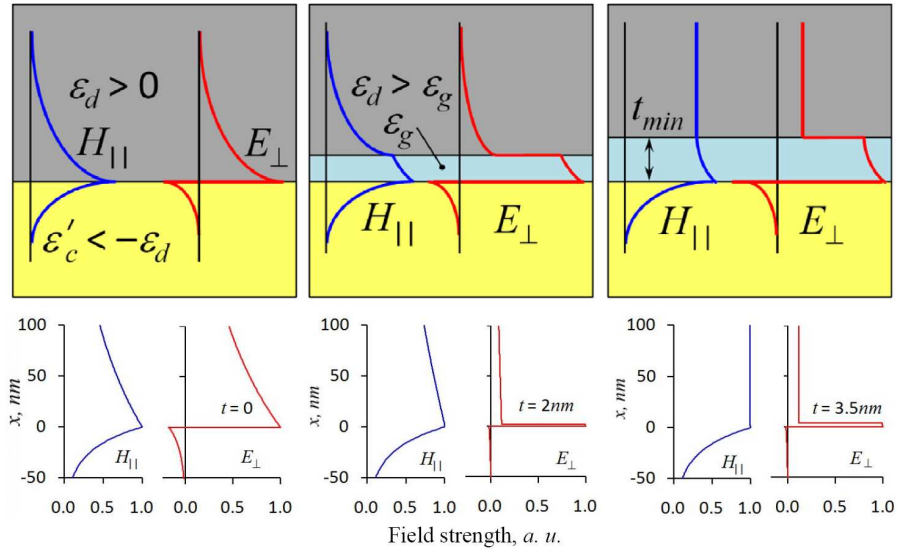


Fig. 2. Graphs at the top show schematically fields of surface plasmon polariton, SPP (left), modified SPP in a CGD structure (center), and critical CGD mode close to cutoff (right). Thickness of the low-index gap layer is shown not to scale compared to the field penetration depth into the high-index dielectric and conductor claddings. Graphs at the bottom are a concrete numerical illustration for the case of $n_d = 4.26$ (Ge), $n_g = 1.44$ (SiO₂), $n_c = 0.583 + 10.1i$ (Au), $\lambda = 1590\text{ nm}$. Thickness of the gap layer is 0 (left), 2 nm (center), and 3.5 nm (right). Blue curves show strength of tangent component of magnetic field, and red curves show normal component of electric field.

To better understand the CGD waveguide system consider the modal index of the system as a function of gap thickness (Fig. 2). A case of large index contrast between the gap and the dielectric is discussed below, while systems with small index contrast will be discussed in detail elsewhere. As previously mentioned, when the thickness of the gap is zero a traditional SPP is found at the interface between the conductor and the dielectric with a modal index, n_{SPP} , given by Eq. (1). As a gap, composed of a material with an index less than the index of the top dielectric layer is introduced, the modal index of the system, n_{CGD} , begins to decrease. This decrease in modal index will continue until the gap thickness t reaches a certain value t_{min} . For $0 < t < t_{min}$, a low loss, CGD mode will exist provided n_{CGD} remains above the index of the top dielectric cladding layer.

In a similar fashion, consider the case when the gap thickness is infinite. In this case a traditional SPP results located at the interface between the conductor and the gap material. In a system with large enough index contrast between the gap and dielectric, the effective index of this SPP is smaller than the index of dielectric cladding. As the gap thickness t is decreased, the modal index of this “large gap” CGD system, or n_{LG-CGD} , begins to increase, but is less than the index of the upper dielectric cladding layer. This represents a high loss mode as the mode shows substantial leakage into the higher index cladding. Due to the leakage, the large-gap mode is not really a well-confined guided mode, but rather a quasi-mode that always shows some net energy flux normal to the interfaces and directed away from the structure. It is not uncommon though: the Otto configuration for excitation of SPP is essentially the large-gap CGD structure. The increase in modal index continues until the thickness reaches a second cutoff thickness of t_{max} . In the region $t_{min} < t < t_{max}$, no physically relevant solution to Eq. (6) exists. Because the LG-CGD mode is a quasi-mode, the second cutoff at t_{max} is not defined as clearly as the real cutoff at t_{min} . When gap thickness approaches t_{max} , leakage losses take on extremely large values, reaching up to 10^5 cm^{-1} for some representative structures. The guided mode is over-damped in this case. Various different definitions of the cutoff for the quasi-mode are possible. For example, one can define cutoff by requiring that the propagation length is long enough to allow for at least one oscillation of the electromagnetic field. This would translate into $4\pi \text{Im}(n^*) = \text{Re}(n^*)$. In this paper, to be consistent with the cutoff definition for t_{min} , we define cutoff t_{max} by setting the real part of the effective index of the LG-CGD mode equal to the refractive index of the dielectric $\text{Re}(n^*) = n_d$.

The existence of two cutoff thicknesses appears to be unique to the CGD system with large index contrast between the gap and the dielectric. In traditional core/cladding waveguide systems, a lower bound to the dimensions of the waveguide geometry is well known. However, in this case, a low loss, bound mode exists only until the thickness increases to a certain value, beyond which there is either no mode or a loosely bound leaky mode. It is also interesting to note that a modal index discontinuity develops in the region $t_{min} < t < t_{max}$. This discontinuity in modal index can be avoided, if the upper, higher index cladding layer is patterned into its own waveguide geometry. This is similar to the situation of Reference [24] – [26], where modal confinement was available in a high index nanowire situated above a conductor. As the spacing between the nanowire and the conductor becomes less than the t_{min} of this work, the CGD mode became the dominant mode and the need for a patterned nanowire was eliminated.

Because the electric field components of TM polarized modes are mainly normal to the interfaces, and because the boundary conditions require continuity of the normal component of electric displacement vector, an enhanced electric field is expected in the gap, with the enhancement factor on the order of ϵ_d/ϵ_g similar the electric field enhancement observed in slot waveguides and for SPPs. The electric field enhancement in this case can also be directly deduced from Maxwell’s equations. For TM- polarized modes, one can find $E_{\perp} \sim H_{\parallel} n_{CGD} / \epsilon$, where $E_{\perp}$ is the normal component of the electric field, $H_{\parallel}$ is the tangent component of the magnetic field, and ϵ is the permittivity of the material at a given point in the cross section of

the waveguide. The $H_{\parallel}$ field, being continuous, is only slightly modified by the presence of the thin gap layer. The $E_{\perp}$ field, in contrast, shows a large increase in the gap due to the $1/\varepsilon$ term. Beyond arithmetic, the physical origin of the electric field buildup in the gap is caused by the surface polarization charges, quite similar to the field enhancement in the slot waveguides. The enhancement of fields as a function of gap thickness is illustrated in Fig. 2 for gap thickness t such that $0 < t < t_{min}$.

Because of the importance of the value of t_{min} to the CGD system, the minimum gap spacing will now be approximated. This is done by letting $n_{CGD} \rightarrow n_d$, in Eq. (7), linearizing Eq. (6), and keeping only its real part. The estimation for the critical thickness of the gap layer is thus found to be:

$$t_{min} \approx \text{Re} \left[\frac{\lambda}{4\pi\sqrt{\varepsilon_d - \varepsilon_g}} \ln \left(\frac{\varepsilon_c \sqrt{\varepsilon_d - \varepsilon_g} - \varepsilon_g \sqrt{\varepsilon_d - \varepsilon_c}}{\varepsilon_c \sqrt{\varepsilon_d - \varepsilon_g} + \varepsilon_g \sqrt{\varepsilon_d - \varepsilon_c}} \right) \right] \\ \approx \frac{\lambda}{4\pi\sqrt{\varepsilon_d - \varepsilon_g}} \ln \left(\frac{\varepsilon'_c \sqrt{\varepsilon_d - \varepsilon_g} - \varepsilon_g \sqrt{\varepsilon_d - \varepsilon'_c}}{\varepsilon'_c \sqrt{\varepsilon_d - \varepsilon_g} + \varepsilon_g \sqrt{\varepsilon_d - \varepsilon'_c}} \right). \quad (8)$$

With $\varepsilon_d \ll |\varepsilon'_c|$, the estimation for the cutoff thickness can further be simplified to

$$t_{min} \approx \frac{\lambda}{2\pi} \frac{\varepsilon_g}{\sqrt{|\varepsilon'_c|}} \frac{1}{\varepsilon_d - \varepsilon_g}. \quad (9)$$

5. Results of numerical simulations

To connect with prior literature, we initially choose gold as the conductor. Introducing now a group-IV plasmonic “theme” for further numerical estimations, we set $n_d = 4.26$ (Ge [27]) and $\varepsilon_c = (0.583 + i \cdot 10.1)^2$ (Au [27]) at the wavelength of $\lambda = 1590$ nm where highly doped n -type germanium is expected to produce optical amplification [28]. In this case, the ratio $\varepsilon_d / |\varepsilon'_c| \approx 0.18$ is not a particularly small number, so we rely on the exact formula (1), which gives $n_{SPP} = 4.692 + i \cdot 0.058$. This corresponds to rather high values of $\text{Re}(n_{SPP}) - n_d = 0.432$ and $\alpha_{SPP} = 4590 \text{ cm}^{-1}$. For comparison, with $n_d = 1.0$, one finds $\text{Re}(n_{SPP}) - n_d = 0.0049$ and $\alpha_{SPP} = 45 \text{ cm}^{-1}$. In this and further numerical examples both dielectric cladding and gap materials are considered lossless. Regarding dielectric loss, as discussed below, the germanium is assumed to be n -doped and pumped appropriately to overcome its inherent loss at 1590 nm.

According to (8), with the gap layer index of $n_g = 1.44$ (SiO_2) the maximum thickness of the gap layer in the Ge/ SiO_2 /Au CGD system is estimated to be 3.5 nm. Replacing the SiO_2 gap with a vacuum of index $n_g = 1.0$ results in a t_{min} of only 1.6 nm. A gap made of Si ($n_g = 3.48$) would result in $t_{min} = 61$ nm. To compare results with Reference [26], setting $n_d = 2.7$ (CdS), $n_g = 1.38$ (MgF_2), and $n_c = 0.13 + 2.88i$ at $\lambda = 489$ nm, one finds an estimated value $t_{min} = 14$ nm. Even though the ratio $\varepsilon_d / |\varepsilon'_c| \approx 0.88$ is no longer a small parameter, the estimated value of t_{min} matches reasonably well with the gap size found from FEM analysis. Exact values of t_{min} , as well as modal index and losses of CGD modes for the gap size in the range $0 < t < t_{min}$ are found by solving (6) – (7) numerically. This explains why the gap should be nano-thin: the modal index of SPP at zero gap is only slightly above the index of dielectric, so it takes only few nanometers of low-index gap to bring it down to the cutoff condition. If the gap material has index comparable to, but still lower than, the refractive index of the dielectric cladding, the gap may be much thicker as in the case of the Ge/Si/Au structure.

Equations (6) – (7) are transcendental and not resolved analytically with respect to the propagation constant of the CGD mode. However, an approximation similar to the one used in deriving (9), that is, neglecting losses, found by linearizing the right-hand and left-hand parts of the equation, as well as assuming the good conductor conditions – allows for solving (6) – (7) for any given gap layer thickness $0 < t < t_{min}$:

$$Re(n_{CGD}) \approx n_d \left[1 + \frac{\varepsilon_d}{2} \left(\frac{\varepsilon_d - \varepsilon_g}{\varepsilon_g} \frac{2\pi}{\lambda} (t_{min} - t) \right)^2 \right]. \quad (10)$$

It is interesting that within these approximations, properties of the conductor affect the cutoff thickness t_{min} , but beyond that the modal index of the CGD mode (10) is mainly defined by the properties of the dielectric cladding and low-index gap. Of course, setting $t = 0$ in (10) and using expression (9) for the cutoff thickness t_{min} , one gets an approximate modal index for the SPP at the dielectric-conductor interface perfectly consistent with (2). It is also convenient to rewrite (10) in the following form:

$$Re(n_{CGD}) \approx n_d + \Delta n_{SPP} \left(1 - \frac{t}{t_{min}} \right)^2, \quad (11)$$

where $\Delta n_{SPP} = Re(n_{SPP}) - n_d$ is defined by (2). When high-index dielectrics are used so that the ratio $\varepsilon_d / |\varepsilon_c'|$ is no longer a small parameter, the accuracy of formula (11) can be improved by using (8) for t_{min} and evaluating Δn_{SPP} based on the exact formula (1):

$$Re(n_{CGD}) \approx n_d + \left[Re \left(\sqrt{\frac{\varepsilon_d \varepsilon_c}{\varepsilon_d + \varepsilon_c}} \right) - n_d \right] \left(1 - \frac{t}{t_{min}} \right)^2. \quad (12)$$

Equations (11) and (12) directly link the CGD mode to the traditional SPP mode so that the CGD is essentially a SPP mode at the high-index dielectric/conductor interface modified by the insertion of a low-index gap layer.

Numerical solutions (discussed below) of (6) – (7) for practically relevant cases indicate that losses for the CGD mode are approximately proportional to $(t_{min} - t)$. Beyond that, careful examination of numerically found dependences $\alpha_{CGD}(t)$ also shows that the second order derivative $d^2\alpha_{CGD}/dt^2$ typically changes from positive values at $t \rightarrow 0$ to negative at $t \rightarrow t_{min}$. The fine details of the $\alpha_{CGD}(t)$ dependence are thus fairly complicated even when t_{min} is much smaller than the vacuum wavelength. Derivation of an approximate formula for the losses of the CGD mode is somewhat cumbersome because one can no longer ignore the imaginary part of the conductor's permittivity ε_c'' , thus, the analysis would have to include two small parameters $\varepsilon_d / |\varepsilon_c'| \ll 1$ and $\varepsilon_c'' / |\varepsilon_c'| \ll 1$. While it is doable, the final expression is too convoluted to be useful. We therefore suggest, without rigorous analytical justification though, that

$$\alpha_{CGD} \approx \alpha_{SPP} \left(1 - \frac{t}{t_{min}} \right), \quad (13)$$

where α_{SPP} is the propagation loss for the SPP at the dielectric-conductor interface that can be evaluated using (3) or found using the exact formula (1). In the latter case one gets

$$\alpha_{CGD} \approx \frac{4\pi}{\lambda} Im \left(\sqrt{\frac{\varepsilon_d \varepsilon_c}{\varepsilon_d + \varepsilon_c}} \right) \left(1 - \frac{t}{t_{min}} \right). \quad (14)$$

In partial justification of (14) we would like to say that it definitely gives correct values of losses for CGD modes at $t \rightarrow 0$ and $t \rightarrow t_{min}$, while linear extrapolation for $0 < t < t_{min}$ is a reasonable guess to certain degree confirmed by numerical simulations.

While the accuracy of the approximate formulas discussed here is limited by the requirement $\varepsilon_d \ll |\varepsilon_c|$, they nevertheless provide useful insight concerning scaling of major characteristics of CGD modes, such as cutoff thickness t_{min} , modal index $Re(n_{CGD})$, and losses α_{CGD} as a function of optical properties of materials involved ε_c , n_g , n_d , and the gap size t .

To illustrate the CGD modes considered above we have solved numerically the dispersion Eq. (6) – (7) and also used the approximate formulas (8), (12) and (14) to find the cutoff thickness of the gap layer t_{min} , modal index $Re(n_{CGD})$, and losses α_{CGD} .

The first structure under analysis was Ge/SiO₂/Au, which is an example of very high index contrast between the dielectric cladding and the gap. Figure 3 left shows modal index (top) and losses (bottom) as a function of the gap layer thickness. The solid lines represent the numerical solution of the exact dispersion equation, while the dashed lines are based on the approximate formulas. While the curves plotted using approximate formulas deviate somewhat from those based on numerical solution of the dispersion equation, the overall accuracy of approximation is acceptable for practical estimations. In particular, the parabola-like dependence of $Re(n_{CGD})$ versus $(t_{min} - t)$ is followed quite well.

We also considered modes in a structure that contains a finite thickness of high-index dielectric followed by the semi-infinite cladding with index equal to that of the gap. The dielectric core waveguide has planar geometry in this case. To be consistent with the previous example, the high-index dielectric was Ge ($n_d = 4.26$). The low-index cladding as well as the gap layer was assumed to be made of silica ($n_g = 1.44$). The conductor was gold ($n_c = 0.583 + 10.1i$). The thickness of the dielectric waveguide layer was chosen to be 250 nm. Figure 3 right shows modal index and loss associated with the CGD, the SPP mode associated with the conductor and an infinitely large gap, and the LG-CGD mode with the dielectric patterned into a waveguide geometry similar to the work of Ref [24] – [26].

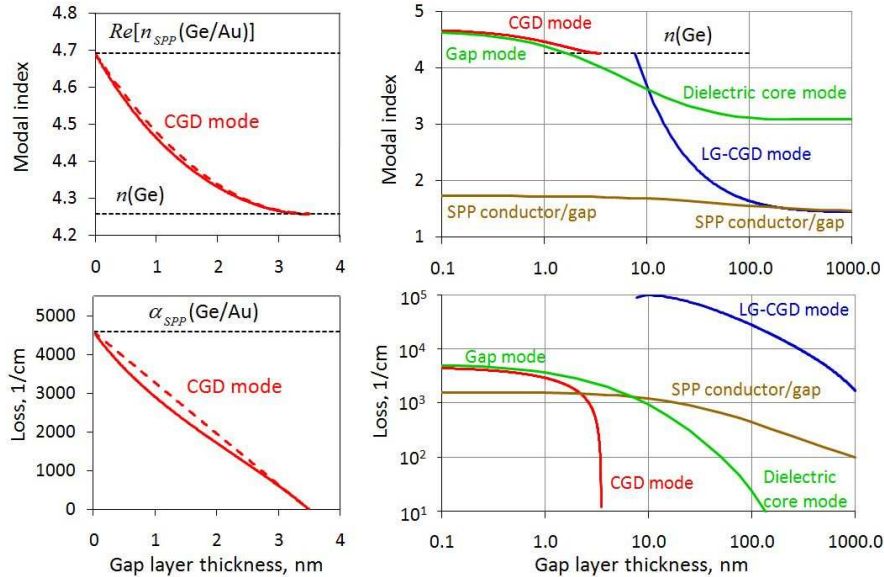


Fig. 3. Modal index (top left) and losses (bottom left) for CGD mode in a structure with $n_d = 4.26(\text{Ge})$, $n_g = 1.44(\text{SiO}_2)$, $n_c = 0.583 + 10.1i(\text{Au})$ at $\lambda = 1590\text{nm}$. Solid lines represent numerical solution of the dispersion equation, and dashed lines calculated using the approximate formulas. Graphs at the right show modal indices (top) and losses (bottom) for CGD mode (red), large gap CGD (blue), dielectric core mode (green), and SPP mode at conductor/gap interface (brown).

Note that a logarithmic scale is used for gap size. The graph at the top shows modal indices, and the graph at the bottom shows modal losses. The red curves correspond to the CGD mode, and they are identical to the solid lines shown at the left part of Fig. 3 except for the logarithmic scale for gap layer thickness. At larger gap values, the same structure supports the large-gap mode LG-CGD (blue curves) which turns into SPP at the gap/conductor interface when the gap thickness is large enough. Here it is clearly seen the modal index discontinuity found for the CGD and LG-CGD modes. Mathematically, when modeling CGD and LG-CGD modes, different branches of one of the complex-valued functions in (7) are used in the dispersion Eq. (6).

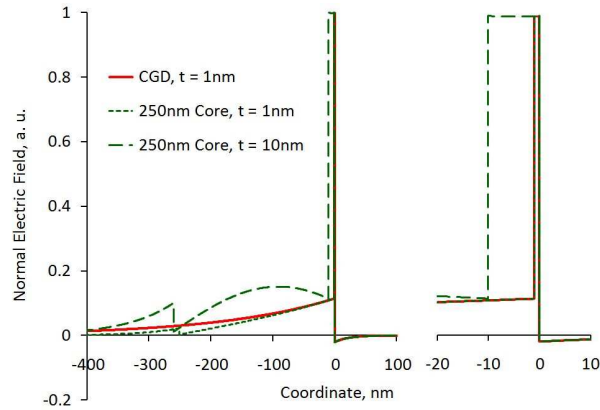


Fig. 4. Mode profiles for Ge/SiO₂(1nm)/Au CGD structure (solid red) and SiO₂/Ge(250nm)/SiO₂(1nm)/Au (short-dash green) and SiO₂/Ge(250nm)/SiO₂(10nm)/Au (long-dash green) dielectric core structures. Graph at the right shows a detail of the profiles in the vicinity of the gap.

The structure with the dielectric core supports two modes. One of them is a traditional SPP mode located at the gap/conductor interface modified by the presence of the high-index dielectric layer (brown curves). The other mode is the mode of the dielectric core at large gap size, which turns into the CGD mode when the gap size is reduced (green curves). There is a range of gap sizes where it is appropriate to call this mode a hybrid of the mode of dielectric core and the plasmon polariton. However, the smaller the gap size, the less hybridization there is. Furthermore, considering modes with strong field confinement in the gap (those with gap layer thickness below t_{min}), the hybridization as such is not needed at all: the high-index layer can be semi-infinite without the need to support a mode of its own. Also noteworthy is that losses for the mode in a structure with a finite thickness of a high-index dielectric core are larger than the losses for the CGD mode for all values of gap thickness from zero to t_{min} . The hybrid mode starts showing smaller propagation loss when the gap size is large, and consequently the tight confinement of energy in the gap is lost.

Figure 4 illustrates the mode profiles for three waveguides: a CGD structure with gap layer thickness of 1 nm and a structure with a 250 nm dielectric core whose gap size is either 1 nm or 10 nm. When the gap size is small (1 nm), the profiles of guided modes in a CGD structure and a dielectric core structure are quite similar. The dielectric core waveguide in this example would show somewhat stronger confinement but would suffer from higher loss. Because the mode profile shows similarity with the exponentially decaying field of plasmonic modes, referring to hybridization with the dielectric core mode would be misleading. On the other hand, with larger gap size (10 nm, which is still small compared to the penetration depth for both SPP and a dielectric waveguide mode), the modal field within the dielectric core shows a local maximum. This resembles the mode of a dielectric waveguide thus justifying the concept of hybridization. A larger field in the dielectric core generally implies less tight confinement of field in the gap.

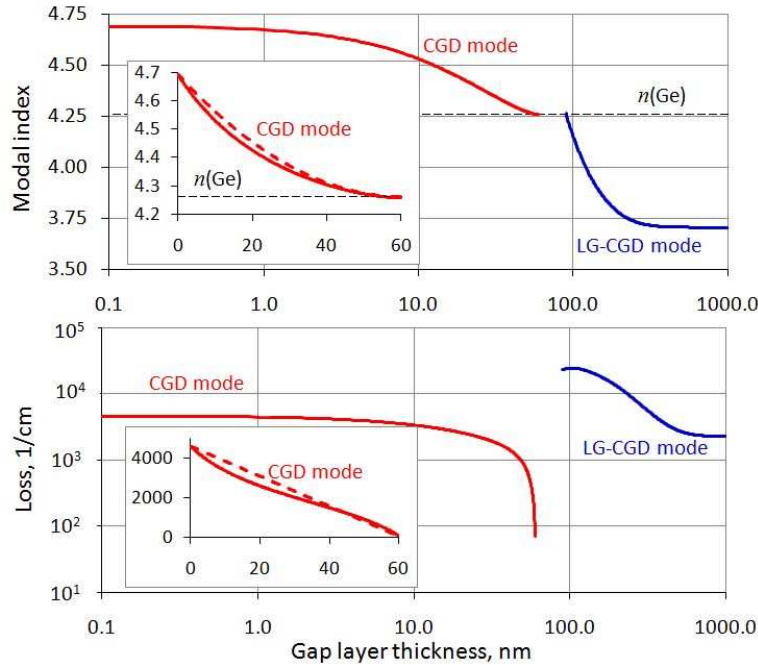


Fig. 5. Modal index (top) and losses (bottom) versus gap layer thickness for CGD (red) and LG-CGD (blue) mode in a structure with $n_d = 4.26(\text{Ge})$, $n_g = 3.48(\text{Si})$, $n_c = 0.583 + 10.1i(\text{Au})$ at $\lambda = 1590\text{nm}$. The insets show modal index and losses for CGD mode in linear scale. Solid curves are calculated by solving numerically the dispersion equation. The dashed lines in the insets are calculated using the approximate formulas.

The above example, with large index contrast, as in Ge/SiO₂ system, is quite illustrative but may not be very practical because it requires the gap to be in the single-nanometer range. We also considered a system with much smaller index contrast: Ge/Si. Figure 5 shows the modal indices (top) and losses (bottom) for Ge/Si CGD system. The same color coding is used as before: the CGD mode is shown in red and the LG-CGD is shown in blue. The insets show CGD modes only using a linear scale for the gap-layer thickness. Solid lines correspond to numerical solution of the dispersion equation, and the dashed lines are calculated using approximate formulas. The cutoff thickness in this case is much larger, 61 nm, but it still provides rather strong confinement. The approximate formulas work equally well for the structure with low index contrast between the dielectric cladding and the gap.

6. Techniques for lateral confinement of the CGD mode

The CGD waveguide is unique in that lateral field confinement is determined by both the conductor and gap thicknesses. Several possible geometries that provide lateral confinement are illustrated in Fig. 6. False colors show light intensity distribution across the structure with red representing the highest intensity and dark blue the lowest. The arrows show the direction of the electric field. In all cases we assumed the dielectric cladding to be semi-infinite to avoid any hybridization. However numerical implementation of this requires an appropriate absorbing layer placed above the dielectric cladding of finite size. The material system is Ge/Si/Au and the simulation is performed using the *COMSOL* commercial software package.

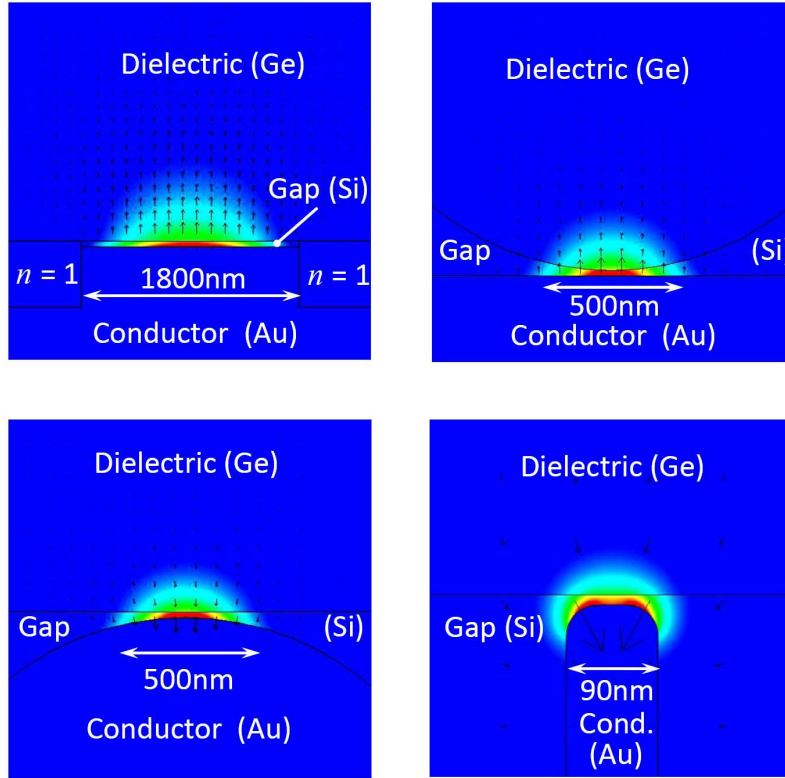


Fig. 6. Sample structures that show lateral confinement. Minimal gap size is 50 nm for the top left, 20 nm top right and bottom left figures, and 10 nm for the bottom right figure. Radius of curved surfaces in the top right and bottom left graphs is 1000 nm. The conductor tip in the bottom right figure has 30 nm wide flat top continued with arcs of 30 nm radius. Colors indicate light wave intensity in the cross section of the structure with dark blue corresponding to minimal intensity and red corresponding to maximal intensity. Arrows show direction of electric field vector.

In general, based on Fig. 6, the vertical size of the mode is approximately equal to the gap height which, in this case, ranges from 10 nm to 50 nm. This corresponds to a modal size in the vertical direction between 0.006 to 0.031λ . One obvious way to provide lateral confinement is to truncate the structure which effectively allows modes only in the regions where the gap thickness is less than t_{min} . The top left graph shows a CGD mode in a structure with a flat high-index dielectric surface, and with the gap size of 50 nm suddenly changing to 500 nm. The wide gap area in this example is assumed to be free space. The narrow-gap region is 1800 nm wide in the lateral direction, and the mode width is about 0.6λ . This geometry allows for reducing the propagation losses to about 700 cm^{-1} . This is in contrast to the previously mentioned SPP mode found at a Ge/Au interface which has propagation losses of 4590 cm^{-1} . As there is practically no field penetration into the wide gap region, the characteristics of the CGD mode in such a structure are not expected to be sensitive to the exact shape of the truncated region. In particular, the width of the wide gap area can be infinitely large.

Another approach is to change the gap size gradually (top right graph). Here the interface between the gap material and the high-index dielectric is a half-cylinder dielectric with radius 1000 nm while the conductor surface is flat. This geometry resembles the structure studied in [26]. Note that the top part of the cylinder is not needed to keep the gap mode in place. The minimum gap size in this illustration is 20 nm and the mode width is about 0.2λ .

The gradual change of the gap size is also achievable with a flat dielectric/gap interface and a curved conductor surface (bottom left) such as a “subterranean” or buried layer of platinum silicide in silicon. As long as the lateral variation of the gap width is the same as in the previous case, the modal field is also very similar. Thus in the bottom left, the mode width is again $\sim 0.2 \lambda$.

Finally, we show that the conductor’s cross section close to the gap/dielectric interface may be actually very small. The bottom right graph illustrates the case in which the conductor has a 30 nm wide flat top with curved side surfaces of 30 nm radii. The gap size here is 10 nm. This is quite an extreme example, far from the basic flat layer geometry considered previously. The confined mode is nevertheless present. Here the mode size is about $0.006 \lambda \times 0.05 \lambda$.

7. Proposed applications

We are visualizing applications for the ultracompact mode sizes reported in this paper. A variety of new CGD-mode devices will be enabled in which the electromagnetic wave energy is confined to lateral dimensions that are significantly smaller than those found in photonic waveguides; that is, the new CGD structure allows for model cross-section dimensions that fall below the diffraction limits of optics.

The two principal applications are the silicon-based plasmon laser, also known as the surface-plasmon laser or SPASER [29] – [31], and ultracompact chem.-bio sensors. Mode propagation losses in the various devices are an issue and will be handled in two ways. First, losses are “moderated” by choosing the gap height to be close to the mode cutoff height as discussed above. Second, increasing the wavelength of operation is a practical strategy. Our prior work on plasmonic channel waveguides [32] – [36] shows clearly that as the wavelength is increased from the near infrared into the midwave and longwave infrared, the loss goes down dramatically and the technology becomes practical.

If we consider for example the far-infrared wavelength region of 50 to 200 μm , the CGD waveguide losses are estimated to be less than 10 dB per cm. That would enable the construction of plasmonic integrated circuits or PLICs which are monolithic, inplane, interconnected networks of various plasmonic channel waveguides. The PLIC comprises waveguided lines, bends, curves, splitters, couplers, combiners, filters and resonators similar in their shape to the waveguides in a photonic integrated circuit (PIC). These PLIC components can be passive or active. A good example of an active component is a device in which one dielectric region is electro-optic – or, more accurately electro-plasmonic, EP – which means the CGD is modified by external electrical control of the semiconductor top layer. Thus, the EP active components will be typically modulators, switches, amplifiers, emitters and detectors.

Regarding the structure of the SPASER, we note that the device can be either optically pumped or electrically pumped for the creation/injection of electrons and holes into the infrared gain-material portion of the spaser structure. There are two good options for locating the gain medium within the device. The first is to fill the gap material with gain. The second is to make the high-index dielectric the gain material. Here are two telecom examples that appear practical. The first is an ultrathin gap consisting of silicon nitride that is heavily doped with 1.55- μm -emissive Erbium ions. The second is a Ge-on-Si dielectric in which the heavily n-doped Germanium is optically excited on its top surface to give strong 1590 nm gain. Additional gain materials will be discussed in a future paper.

8. Conclusions

We have found that the conductor-gap-dielectric structure supports extremely confined guided modes (sub-diffraction size in both the x and y directions) that are SPPs at the conductor-dielectric interface modified by the presence of an ultrathin, low-index gap layer. The electric field of the CGD mode is significantly enhanced in the gap layer, drops sharply at the gap

interface, and decays exponentially into the conductor and dielectric claddings. We derived dispersion equations for the CGD modes and provided an analytical approximation for the cutoff thickness of the gap layer. The CGD mode exists when the thickness of the gap layer is below the cutoff thickness. We also found approximate analytical solutions for the modal index and losses when the gap layer thickness is anywhere between zero and its cutoff value. We confirmed the validity of the approximate formulas by solving the dispersion equation numerically. We suggested possible geometries that provide lateral confinement for the CGD modes and verified this by numerical simulations. The ultrasmall CGD modes will open up new applications in active/passive plasmonic devices and chip-scale systems.

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